

Haotian Gong

Website: ht-gong.github.io
Github: github.com/ht-gong

Email: haotiangong@hotmail.com
Mobile: +1-778-885-6840

EDUCATION

- **University of Michigan** Ann Arbor, United States
PhD Student – Advisors: Lin Ma, Barzan Mozafari; GPA: 4.00/4.00 September 2024 - Expected 2029
- **University of British Columbia** Vancouver, Canada
B.Sc. (Honors) in Computer Science; GPA: 91.4/100 September 2019 - June 2024
Thesis: Less is More: Designing a Flexible Graph Store with Key-values

PUBLICATION

Jialong Li, **Haotian Gong**, Federico De Marchi, Aoyu Gong, Yiming Lei, Wei Bai, and Yiting Xia. 2024. Uniform-Cost Multi-Path Routing for Reconfigurable Data Center Networks. *In Proceedings of the ACM SIGCOMM 2024 Conference (ACM SIGCOMM '24)*. 433–448.

EXPERIENCE

- **UMich Database Group, University of Michigan** Ann Arbor, USA
PhD Student August 2024 - Present
 - Designing a GPU-accelerated OLAP database in PyTorch, enabling terabyte-scale analytics workloads to run on GPUs. Publication in preparation for VLDB 2025.
- **ByteDance** Beijing, China
Developer Intern - Ads Monetization Team June 2024 - August 2024
 - Designed and maintained Apache Hive data pipelines and Apache Flink streaming SQL jobs for real-time advertisement platform monitoring, supporting processing of up to 1M queries per second.
- **Max Planck Institute for Informatics** Saarbrücken, Germany
Research Intern - Prof. Yiting Xia May 2023 - September 2023
 - Co-designed a novel routing algorithm for reconfigurable optical data center networks, achieving up to 51% reduction in flow completion time compared to state-of-the-art solutions, resulting in second-author publication at SIGCOMM.
- **Systopia Lab, The University of British Columbia** Vancouver, Canada
Research Intern - Prof. Margo Seltzer May 2022 - April 2023
 - Contributed C++ code to the “Flexograph” graph processing system; leveraged the OpenMP parallel programming framework to speed up graph algorithms by up to 14×. [\[Code\]](#)
 - Independently explored 2 paradigms of memory - storage interactions in graph processing systems. Presented findings through written and verbal reports in graduate seminar course. [\[Report\]](#)
- **Digital Health Innovation Lab, BC Hospitals** Vancouver, Canada
Software Developer May 2021 - August 2021
 - Built a Python pipeline that modularizes and automates patient data ingestion into the i2b2 data warehouse, reduced size of legacy SQL codebase by 95%.
- **Philips** Vancouver, Canada
DevOps Engineer Intern January 2021 - April 2021
 - Prototyped Slack chatbot using a full serverless architecture that pushes AWS CloudWatch alerts to Slack channels, eliminating manual monitoring effort for the DevOps team.

SERVICE

- NSDI Artifact Evaluation Committee 2025

TECHNICAL PROJECTS

- **ED Forecast (2023)** Selected student project for Borealis AI “Let’s Solve it” challenge. Implemented a time-series model that forecasts patient arrival patterns for emergency departments in hospitals around Canada. [\[Code\]](#)
- **Course Copilot (2023)** Prototype of a Retrieval-Augmented Generation student assistant for higher education courses. Top 10% of applications to 2023 YC batch. [\[Report\]](#)

SKILLS SUMMARY

- **Languages:** C++, C, Python, Java, SQL
- **Libraries & Frameworks:** PyTorch, CUDA, OpenMP, Kubernetes, Django
- **Certificates:** AWS Certified Solutions Architect – Associate

HONORS AND AWARDS

- University of Michigan First Year PhD Fellowship 2024
- EPFL PhD Fellowship (declined) 2024
- UBC Dean of Science Scholarship 2023
- UBC Science Undergraduate Research Experience Award 2022
- UBC Trek Excellence Scholarship, awarded to top 5% in faculty by sessional GPA 2020, 2021, 2022
- UBC Basketball Intramurals Champions 2022